



BITS Pilani
Pilani Campus



SURVEYING (CE F213)

(Computations of Areas and Volumes)

Computations of Areas and Volumes is one of the objectives in many surveys. After plotting and finishing the plan, the surveyor need to calculate the areas and volume of earthwork etc. in many engineering projects.

Computations of Areas

- (i) When land is bought and sold, measurement of area is necessary.
- (ii) Area of reservoir is used to calculate the evaporation loss.
- (iii) Area within the irregular boundaries in many cases of land survey.

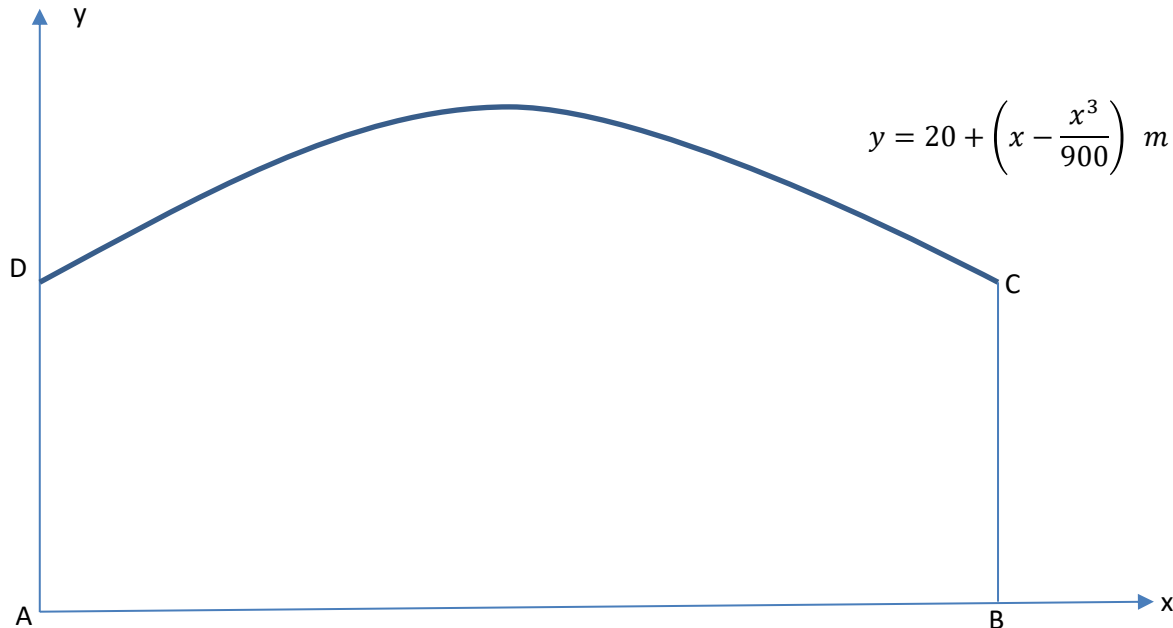
Various method of calculation of areas are

- (i) Trapezoidal rule,
- (ii) Simpson's $1/3$ rule (no of divisions should be multiple of 2)
- (iii) Simpson's $3/8$ rule (no of divisions should be multiple of 3)
- (iv) Average ordinate rule
- (v) Co-ordinate method
- (vi) Mid- ordinate rule

Do not ignore any ordinate if that is zero also, in the above methods. If any method is not directly applicable due to the number of division constraint, then calculate the area of the extra division by Trapezoidal rule. Apply the given method for the rest of the divisions.

Computations of Areas and Volumes

Q1: In the given figure, the property(ABCD) has curved boundary CD. The equation of the curved boundary is mentioned in the figure. Side AB is on x-axis and side AD is on y-axis. Side BC is parallel to side AD. The length of AB is 30m. Find the area of the property by **all methods** by dividing side AB in 8 equal divisions. Find the actual area and the percentage difference of these results with the actual area.



We have

$$y = 20 + \left(x - \frac{x^3}{900} \right) m$$

AB = 30 m,

8 divisions has to be made

So,

$$d = 30/8 = 3.75m$$

$$y_0 = y_1 = 20$$

$$y_{3.75} = y_2 = 20 + \left(3.75 - \frac{3.75^3}{900} \right) = 23.691$$

$$y_{7.5} = y_3 = 27.031$$

$$y_{11.25} = y_4 = 29.667$$

$$y_{15} = y_5 = 31.25$$

$$y_{18.75} = y_6 = 31.425$$

$$y_{22.5} = y_7 = 29.843$$

$$y_{26.25} = y_8 = 26.152$$

$$y_{30} = y_9 = 20$$

$$\begin{aligned}\text{Area by trapezoidal rule} &= 3.75 \left(\frac{20+20}{2} + 23.691 + 27.031 + 29.667 + 31.25 + 31.425 + 29.843 + 26.152 \right) \\ &= 821.471 \text{ m}^2 \quad [5]\end{aligned}$$

$$\begin{aligned}\text{Area by Simpson's rule} &= \frac{37.5}{3} (20 + 20 + 4(23.691 + 29.667 + 31.425 + 26.152) + 2(27.031 + 31.25 + 29.843)) \\ &= 824.985 \text{ m}^2 \quad [5]\end{aligned}$$

$$\text{Actual area} = \int_0^{30} 20 + \left(x - \frac{x^3}{900} \right) dx = \int_0^{30} 20 dx + \int_0^{30} \left(x - \frac{x^3}{900} \right) dx = 825 \text{ m}^2 \quad [5]$$

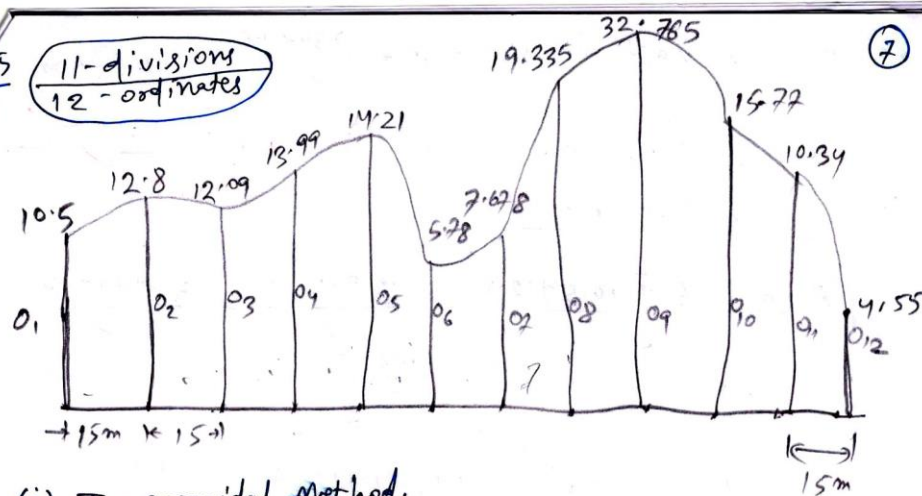
$$\% \text{age error} = \frac{825 - 821.471}{825} \times 100 = 0.427\% \text{ (For trapezoidal rule)} \quad [1]$$

$$\% \text{age error} = \frac{825 - 824.985}{825} \times 100 = 0.0018\% \text{ (For Simpson's rule)} \quad [1]$$

Q2: The offsets taken at 15m interval from the irregular boundary to a chain line are 10.5m, 12.8m, 12.09m, 13.99m, 14.21m, 5.78m, 7.678m, 19.335m, 32.765m, 15.77m, 10.34m and 4.55m. Find the area between the chain line and the boundary by Trapezoidal method, Simpson's $1/3$ rule, Mid ordinate method and Average ordinate method.

15

11-divisions
12-ordinates



(i) Trapezoidal method.

$$\begin{aligned}
 \text{Area} &= d \left[\frac{O_1 + O_{n+1}}{2} + O_2 + O_3 + \dots + O_n \right] \\
 &= \frac{15}{2} \left[\frac{10.5 + 4.55}{2} + 12.8 + 12.09 + \dots + 10.34 \right] \\
 &= 2284.245 \text{ m}^2
 \end{aligned}$$

(ii) Simpson's $\frac{1}{3}$ rule. In this case even number of ordinates are there. So we can't apply this Simpson's $\frac{1}{3}$ rule directly. So we evaluate the area of 1st 10 divisions by Simpson's $\frac{1}{3}$ rule. Last divn area by trapezoidal rule.

Area of 1st 10 div's

$$\begin{aligned}
 \text{Area} &= \frac{d}{3} \left[\text{1st + last ordinate} + 4(\text{sum of even ordinates}) + 2(\text{sum of odd ordinates}) \right] \\
 &= \frac{15}{3} \left[10.5 + 10.34 + 4(12.8 + 13.99 + 5.78 + 19.335 + 15.77) + 2(12.09 + 14.21 + 7.678 + 32.765) \right] \\
 &= 2125.13
 \end{aligned}$$



$$\text{Area of last division} = 15 \times \frac{10.34 + 4.55}{2} = 111.675 \text{ m}^2$$

$$\text{So, total area} = 2125.13 + 111.675 = \underline{2236.805 \text{ m}^2}$$

(iii) Mid ordinate method.

$$\text{Area} = 15 \left[\frac{10.5 + 12.8}{2} + \frac{12.8 + 12.09}{2} + \frac{12.09 + 13.99}{2} + \dots + \frac{10.34 + 4.55}{2} \right]$$

$$= 15 [11.65 + 12.445 + 13.04 + 14.1 + 9.995 + 6.729 + 13.5065 + 26.05 + 24.2675 + 13.055 + 7.445]$$

$$= \underline{2284.245 \text{ m}^2}$$

(iv) Average ordinate method.

$$\text{Area} = 11 \times 15 \times \left[\frac{o_1 + o_2 + \dots + o_{12}}{12} \right]$$

$$\text{Area} = 11 \times 15 \times 13.31733$$

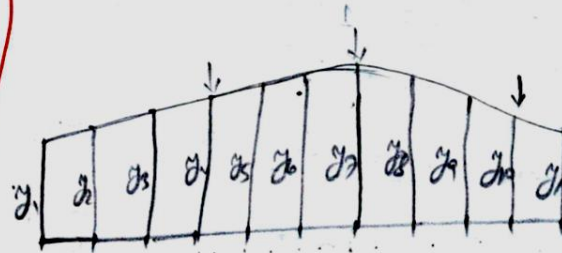
$$= \underline{2197.36 \text{ m}^2}$$

Q3: Four chain lines AB, BC, CD and DA are located on the ground, with co-ordinates of A(20,20), B(40,80), C(100,90) and D(75,30). Along the external side of line AB, the offsets of an irregular boundary are 0,2,1.5,3.8,2.1,2.5,1.99,3.1,1.77,2.05 and 0. The 1st offset is at A and the last offset is at point B. The offsets are at equal intervals. Find the area enclosed among the irregular boundary AB and other three chain lines. While calculating the area between the irregular boundary AB and chain line AB use (i)Trapezoidal method, (ii) Simpson's $\frac{1}{3}$ rule, and (iii) Average ordinate method. All dimensions are in m.

Q4: The boundary of a patch of land from a chain line is given as a function as $y = 10 + 0.1x - 10^{-5}x^3$. The length of chain line is 100m. Find the area of the land using (i) Trapezoidal rule, (ii) Simpson's 1/3 rule, (iii) Simpson's 3/8 rule, (iv) Average ordinate rule and (v) co-ordinate method, by dividing the chain line in 10 divisions. Check the result with exact integration.

No. 3 $y = 10 + 0.1x - 10^{-5}x^3$ ← is the boundary
(to find Area)

$$\begin{aligned} y_1 &= 10 \\ y_2 &= 10.99 \\ y_3 &= 11.92 \\ y_4 &= 12.73 \\ y_5 &= 13.36 \\ y_6 &= 13.75 \\ y_7 &= 13.84 \\ y_8 &= 13.57 \\ y_9 &= 12.88 \\ y_{10} &= 11.71 \\ y_{11} &= 10 \end{aligned}$$



$$d = 10 \text{ m}$$

(i) Trapezoidal Rule.

$$\begin{aligned} \text{Area} &= d \left[\frac{y_1 + y_{11}}{2} + y_2 + y_3 + \dots + y_{10} \right] \\ &= 10 [10 + 114.75] = 1247.5 \text{ m}^2 \end{aligned}$$

(ii) Simpson's $\frac{1}{3}$ Rule (we need even no of divisions) here it 10 div's
So ok.

$$\begin{aligned} \text{Area} &= \frac{d}{3} [(y_1 + y_{11}) + 4(y_2 + y_4 + y_6 + y_8 + y_{10}) \\ &\quad + 2(y_3 + y_5 + y_7 + y_9)] \\ &= \frac{10}{3} [20 + 4(62.75) + 2(52)] \\ &= 1250 \text{ m}^2 \end{aligned}$$

(iii) Simpson's $\frac{3}{8}$ Rule. (we need: no of div's should be multiple of 3)

$$\begin{aligned} \text{Area} &= \frac{3d}{8} [(y_1 + y_{10}) + 3(y_2 + y_3 + y_5 + y_6 + y_8 + y_9) \\ &\quad + 2(y_4 + y_7)] = \text{Area of 1st 9 div's} \end{aligned}$$

$$= \frac{3 \times 10}{8} [(10 + 11.71) + 3(10.99 + 11.92 + 13.36 + 13.75 + 13.57 + 12.88) + 2(12.73 + 13.84)]$$

$$= \frac{30}{8} [21.71 + 3 \times 70.47 + 2 \times 26.57]$$

$$= 1140.925 \text{ m}^2$$

$$\text{Area of last (10th) div} = 10 \times \left(\frac{11.71 + 10}{2} \right) = 108.55 \text{ m}^2$$

$$\text{Total Area} = 1249.525 \text{ m}^2 \quad (2)$$

(iv) Average estimate Rule

$$(1) \text{ Area} = 100 \times \left[\frac{y_1 + y_2 + \dots + y_n}{n} \right] = 100 \times (12.25) = 1225 \text{ m}^2$$

(v) Co-ordinate Method

$$\begin{array}{cccccccccccccccc} x: & 0 & 10 & 20 & 30 & 40 & 50 & 60 & 70 & 80 & 90 & 100 & 100 & 0 \\ y: & 0 & 10.99 & 11.92 & 12.73 & 13.36 & 13.75 & 13.84 & 13.57 & 12.88 & 11.75 & 10 & 0 & 0 \end{array}$$

$$\begin{aligned} \sum P &= 10 \times 11.92 + 20 \times 12.73 + 30 \times 13.36 + 40 \times 13.75 + 50 \times 13.84 \\ &+ 60 \times 13.75 + 70 \times 12.88 + 80 \times 11.75 + 90 \times 10 \\ &= 5583.2 \end{aligned}$$

$$\begin{aligned} \sum Q &= 10 \times 10 + 10.99 \times 20 + 11.92 \times 30 + 12.73 \times 40 + 13.36 \times 50 + 13.75 \times 60 \\ &+ 13.84 \times 70 + 13.57 \times 80 + 12.88 \times 90 + 11.75 \times 100 + 10 \times 100 \\ &= 8068.2 \end{aligned}$$

$$\begin{aligned} \text{Area} &= \left| \frac{1}{2} (\sum P - \sum Q) \right| = \left| \frac{1}{2} (5583.2 - 8068.2) \right| \\ &= 1242.5 \text{ m}^2 \quad (1) \end{aligned}$$

$$\text{Integration} \int_0^{100} (10 + 0.1x - 10^{-5}x^3) dx = 1250 \text{ m}^2 \quad (1)$$

